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① Plaintext : MEETATNOON

Key : 7.

Numeric values : A = 0, B = 1, C = 2 .... Z = 25

Ceaser cipher.

A	B	C	D	E	F	G	H	I
0	1	2	3	4	5	6	7	8
J	K	L	M	N	O	P	Q	R
9	10	11	12	13	14	15	16	17
S	T	U	V	W	X	Y	Z	
18	19	20	21	22	23	24	25	

Encryption

$$C = (P + 7) \bmod 26.$$

$M = (12 + 7) \bmod 26 = 19$

$E = (4 + 7) \bmod 26 = 11$

$T = (19 + 7) \bmod 26 = 0$

$A = (0 + 7) \bmod 26 = 7$

$T = (19 + 7) \bmod 26 = 0$

$N = (13 + 7) \bmod 26 = 20$

@: 19 = T , 11 = L, D = B, 7 = H, 21 = V, 20 = U.

Ciphertext = T L L B H B U V V U.

Decryption:

$$p = (c - 7) \bmod 26$$

T = (19 − 7) mod 26 = 12

L = (11 − 7) mod 26 = 4

B = (0 − 7) mod 26 = 19

H = (7 − 7) mod 26 = 0

U = (20 − 7) mod 26 = 14

V = (21 − 7) mod 26 = 13.

Vegenerance cipher.

Plaintext : MEETATNOON

Key SUNSUNSUNS.

1.

P : M	E	E	T	A	T	N	O	O	N
K : S	U	N	S	U	N	S	U	N	S
Calculate : 12+18	20+4	4+13	19+18	0+20	19+13	13+18	14+20	13+14	13+18
=30	=24	=17	=37	=20	=32	=31	34	=27	31
mod 26 : 4	24	17	11	20	6	5	8	1	5

CODE

cipher text : E Y R L U G F I B F

2.

$$K = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}$$

Plaintext: MATH

$MA = [12 \quad 0]$

$TH = [19 \quad 7]$.

$C = K \times P \mod 26$.

For MA,

$$C = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \begin{bmatrix} 12 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 36 \\ 60 \end{bmatrix} \mod 26$$

$$= \begin{bmatrix} 10 \\ 8 \end{bmatrix}$$

$10 = R$, $1 = B$.

For TH,

$$C = \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix}^T \begin{bmatrix} 19 \\ 7 \end{bmatrix}$$

$$= \begin{bmatrix} 87 + 14 \\ 95 + 49 \end{bmatrix} = \begin{bmatrix} 101 \\ 144 \end{bmatrix} \bmod 26$$

$$= \begin{bmatrix} 23 \\ 14 \end{bmatrix}$$

Determinant

$$K = (3 \times 7) - (2 \times 5)$$

$$= 21 - 11 = 11$$

$\gcd(11, 26) = 1$

$11 \times 19 = 209$

$209 \bmod 26 = 1$

$11^{-1} = 19$

1 87 14 101

29 3 87 4 19 5 95 95 49 144 5 26 5 130 14 2 26 9 104 1 26 3 78

4 11 101 78 23

$$K^{-1} = (K)^T \text{ adj } K \mod 26$$

$$= 19 \begin{bmatrix} 3 & 2 \\ 5 & 7 \end{bmatrix} \mod 26.$$

$$= \begin{bmatrix} 57 & 38 \\ 95 & 133 \end{bmatrix} \mod 26$$

$$= \begin{bmatrix} 5 & 12 \\ 17 & 3 \end{bmatrix}$$

Determinant

$$(5 \times 3) - (17 \times 12)$$

$$= 15 - 204 = 189.$$

$$GCD(-189, 26) = 0.$$

$$\therefore K \times K^{-1} \mod 26$$

$$= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

15+51
57

2
19
3
57
1
19
2
38
2
19
5
95
6
19
7
133
1
26
2
52
1
38
26
12
1
17
12
34
1
26
17
3
120 414
15
78
1
89
2
26
4
104
1
26
2
52
3
26
5
130
4
26
7
182
2
17
3
51
10
21

Ex. Let us take

Plain text : MEET.

key : 3

a) Ceaser cipher:

Encryption :

$$P = (C + K) \bmod 26.$$

$M(12) = (12+3) \bmod 26 = 15$

$E(4) = (4+3) \bmod 26 = 7$

$T(19) = (19+3) \bmod 26 = 22.$

15 = P , 7 = H , 22 = W.

Ciphertext : P H H W.

Decryption :

$$C = (P - K) \bmod 26.$$

Plaintext = M E E T.

b) playful cipher :

Plaintext : MEET

~~Key : REXX~~

- Make 5×5 matrix.
- Both letter has seperate ~~word~~ letter.
- Take I & J as same

M	O	N	A	R
C	H	Y	B	D
E	F	G	I	K
L	P	Q	S	T
U	V	W	X	Z

For ME,

ME = CL

For ET,

ET = KL.

ciphertext: 'CLKL.

c) Vigenere cipher.

plaintext : MEET

key : KEYK.

~~MAAK~~ =

P :	M	E	E	T
K	K	E	Y	K.
cal:	12+10	4+4	4+24	19 +10
	=22	=8	=28	=29
mod26:	22	8	2	3

~~Ciphertext:~~

22 = W , 8 = I , 2 = C , 3 = D

ciphertext = W I C D